

Silent Valley (Rural / Mountain / Recon & Stealth)

Executive Summary

This vision is a comprehensive requirements blueprint built for stealth operations, autonomous search, battlefield movement, and evasion across terrain-based elevation. The terrain features forested slopes, normal mountain slopes, abandoned farmsteads, rocky outcrops, small filled valleys, cliffside trails, and hidden observation points. The map emphasizes (Shadow Protocol) autonomous recon logic, sensor fusion under mountain fog and below valleys, canopy relay warfare, and tactical across dense, unseen terrain. In AR/VR/LiteAR, the map pushes at "4-6 m" in passive distance counter's 13000 m physical area representing "1 km" of rural/mountain terrain. The feature includes stealth corridors, elevation-driven detection, concealed movement routes, and recon-focused assessments.

Design Goals & Gameplay

Stealth-Focused Mountain Recon

Quiet valleys and hidden valleys, trails, and infiltration. Forest cover, tree canopies, and elevation shifts create natural blind zones.

Autonomous Search & Terrain-Aware Pathfinding

Logic requires recon logic, sensor fusion, and hidden trails. Autonomous logic selects routes based on concealment, elevation advantage, and sensor clarity.

Concealment & Ambush Warfare

Tree, rock, hill, and small mountain slopes create ambush, ambushes, flanking, and sudden close-range engagements dominate.

Sensor & Information Warfare

Fog, humidity, vegetation density, thermal distortion, and mountain occlusion degrade sensors. Relay towers on ridges restore clarity. Destroying relay creates blind pockets across the valley.

Logistics & Resource

Food packs, batteries, and mountain cabins serve as resupply hubs. Gameplay must navigate narrow roads and avoid ambush-prone forest corridors.

Deployment & Progress

A full deployment budget ("10,000 points) forces players to choose between stealth infantry, recon drones, or mountain-adjacent armor. Blue display from the southeast rural farmstead, Red display from the northwest ridge-top operational cabin.

Spatial Constraints (Nreal AR)

Scene Capabilities
Near light AR: 1.5° diagonal FOV, 1800x1080 per eye resolution. Optimal viewing distance "4-6 m".

Playable Area & Scaling

10x10 m physical area = 1 km² rural/mountain zone (0.000 scales). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking allows horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render past real-world ones, ensure minimal real-world occlusion.

Layout Overview

Coordinate System

Key Locations

Blue HQ (Forward Recon Camp) -- (0,0), 100x100 m. Basic, crates, recon drone, rural station.

Red HQ (Hidden Recon Camp) -- (1000,1000), 100x100 m. Disguised as abandoned cabin with radio tower, thermal release, and camouflage.

Forward Recon Cabin -- (200,100), 100x50 m. One window, "Secret room" inside (radio cabinet, stealth drone control).

Abandoned Farmstead -- (700,800), 150x100 m. Basic, abandoned, "Secret room" inside.

Mountain Cabin -- (800,100), 100x50 m. Small, hidden, "Secret room" inside.

South Center -- (500,500), 100x100 m. Recon control room, "Secret room" inside.

Mountain Trail Road -- (100,1000), 100x100 m. One door, "Secret room" inside.

Blue Ridge A -- (100,100), 100x100 m. One door, "Secret room" inside.

Blue Ridge B -- (100,100), 100x100 m. One door, "Secret room" inside.

Secret Forward Outpost -- (100,100) Recon mission area.

Chokepoints & Sightlines

South Center

Secret visibility box, ideal for stealth ambush and sensor disruption.

Ridge Network

Secret, hidden, ideal for stealth ambush and sensor disruption.

Valley Basin

One ridge ridge-prone, broken by vegetation, fog, reduce accuracy.

Mountain Trail Road

One ridge ridge-prone, broken by vegetation, fog, reduce accuracy.

Sightlines

Blue Ridge A includes long-range sensor coverage ("1000 m). Fog and canopy shadows create fog-of-war pockets.

Deployment Zones

Blue Deployment: 100x100 m. Recon station.

Red Deployment: 100x100 m. Recon station.

Secret Forward Outpost: 100x100 m. Recon station.

Assets & Environment

Terrain

Forest, hills, valleys, mountain trails, rural cabins.

Vegetation

Forest, hills, valleys, mountain trails, rural cabins.

Buildings & Props

Forest, hills, valleys, mountain trails, rural cabins.

Effects

Fog, humidity, rain, rain, canopy shadows, heat.

Audio

Wind, leaves, distant machinery, rain, rain, radio chatter.

LOD & Budget

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